Karan Gupta

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Education

University of Waterloo Expected June 2030

Bachelor of Applied Science in Computer Engineering, Minor in Economics ***GPA**: 3.95*

Experience

Overlap (YC S24) Sep 2026 – Present

Software Engineering Intern *San Francisco, CA*

- Enabled data-driven A/B testing by building an embedding-based semantic search pipeline across 20+ internal and external performance metrics, allowing AI agents to retrieve relevant signals for content optimization
- Built a file-size-capped 4 FPS video proxy with motion-adaptive Gemini frame sampling, speeding up scene detection 2.4× and enabling AI clip generation on long-form videos spanning 2–7 hours

Amazon May 2026 – Aug 2026

Software Development Engineering Intern *Toronto, ON*

- Decomposed a 9,400-line warehouse decision engine into pluggable microservices, enabling dynamic selection of item-picking optimization algorithms deployed across every fulfillment center
- Reduced legacy service migration time 98% (2 weeks → 2 hours) by building an agentic AI migration system with autonomous architecture discovery and 20 reusable deterministic transformation skills
- Re-architected the execution engine across ECS, Lambda, Step Functions, and S3, eliminating redundant computation in algorithm selection and reducing end-to-end execution time from 16s to 4.5s
- Built a semantic-diff validation harness to replay production JSON payloads against generated endpoints, evaluating behavioral parity via a confidence metric that flags inconclusive noise across 42 execution paths

Wat.ai (TRACE Subteam) May 2026 – Present

AI Software Engineer, Evaluation Lead *Waterloo, ON*

- Architected TRACE's deterministic agent-evaluation layer by defining Pydantic verifier contracts for pass/fail, evidence, severity, and affected trace steps across attribution, regression testing, and release gates
- Achieved 100% evaluation accuracy across a 25-task policy suite by validating agent execution traces against retrieved documents, expected actions, tool side effects, and final state to detect unsafe or ungrounded behavior
- Built agent guardrail evaluations measuring replay reliability and classifying recovery into 6 outcome categories

Manulife Financial Corporation January 2026 – April 2026

Software Engineering Intern *Waterloo, ON*

- Processed 500K+ Salesforce records per export by building a scalable Python pipeline with Redis caching to extract, transform, and adapt application data into downstream-compatible formats
- Reduced system downtime risk by building a New Relic telemetry exporter and SQL-based monitoring pipeline, enabling early detection of 3+ resource bottlenecks before service degradation

Projects

DominIQ (Hack The North 2026 Winner) | *Python, PyMAVLink, YOLOv8, ArduPilot, OpenCV, WebXR, TypeScript*

- Coordinated 4 heterogeneous autonomous assets by building a concurrent MAVLink mission controller with dynamic search, interception, target confirmation, tracking, and lost-target recovery
- Achieved 98% vessel detection accuracy by fine-tuning YOLO26m and fusing camera detections with robot altitude, orientation, and camera geometry to localize targets in a shared coordinate system without relying on GPS
- Built a Meta Quest 3 interface via WebXR and Meta IWSDK, displaying telemetry, camera feeds, and targets in 3D

Reparo (Hack Canada 2026 Winner) | *Python, Gemini API, React, Node.js, Swift, PyTorch*

- Classified products and assessed repairability with 90%+ accuracy via a Gemini vision-powered agentic AI
- Reduced user search time for replacement parts by 70% and enabled access to 1,000+ real-time listings by integrating Shopify Storefront API and SerpAPI to automate matching and streamline repair-to-checkout flow
- Won 1st place in Reactiv Track ($5,000) at Hack Canada 2026; shortlisted for Most Complex AI Hack

Mr. Clean | *Python, PyTorch, MuJoCo, ROS 2, OpenCV, Stable-Baselines3, Docker*

- Achieved 9/9 collision-free navigation routes with 0 falls and under 10 cm / 5° docking error by building an A* planner with obstacle inflation, spline trajectories, LiDAR/odometry, and closed-loop control
- Achieved 95% pickup success with a MuJoCo-trained reinforcement-learning policy for household objects
- Implemented an Action Chunking Transformer that fuses 3 camera streams with robot state through ResNet-18, a CVAE encoder, and transformer to generate 32-step manipulation trajectories

Skills

Languages: Python, Java, C, C++, Swift, TypeScript, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: PyTorch, TensorFlow, React, Node.js, Django, FastAPI, OpenCV, ROS 2, MuJoCo, Stable-Baselines3, Redis, JUnit, Mockito, Spring, Next.js

Developer Tools: AWS (CDK, ECS, Lambda, S3), Git, Docker, Kubernetes, Raspberry Pi, Arduino, Linux